# Riyazi Online (Unit 10 MCQS)

1. What is the distance between points $A(x_1, y_1)$ and $B(x_2, y_2)$?
   (a) $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
   (b) $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
   (c) $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
   (d) $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

2. The Fundamental Identity of Trigonometry is:
   (a) $\sin^2\theta + \cos^2\theta = 1$
   (b) $\tan\ \theta = \frac{1}{\cot\theta}$
   (c) $\sec\ \theta = \frac{1}{\cos\theta}$
   (d) $\csc^2\theta - \cot^2\theta = 1$

3. Which identity is derived directly from the distance formula in coordinate geometry?
   (a) Midpoint formula
   (b) Pythagorean identity
   (c) Slope formula
   (d) Gradient theorem

4. If $\sin\ \theta = \frac{3}{5}$, what is $\cos\theta$ assuming $\theta$ is in the first quadrant?
   (a) $\frac{4}{5}$
   (b) $\frac{3}{4}$
   (c) $\frac{5}{3}$
   (d) $\frac{1}{2}$

5. The coordinates of a point 5 units away from origin making angle $\theta$ with x-axis are:
   (a) $(5\cos\ \theta, 5\sin\ \theta)$
   (b) $(5\theta, 5\theta)$
   (c) $(5\cos\ \theta, 5\sin\ \theta)$
   (d) $(x^2, y^2)$

6. Which trigonometric identity reflects the Pythagorean Theorem?
   (a) $\sec^2\theta - \tan^2\theta = 1$
   (b) $\sin^2\theta + \cos^2\theta = 1$
   (c) $\tan\ \theta = \frac{\sin\theta}{\cos\theta}$
   (d) $\cos^2\theta = 2\cos^2\theta - 1$

7. Which function is always positive in the first quadrant?
   (a) Only sin
   (b) Only tan
   (c) Only cos
   (d) All functions

8. Which of the following is correct if $\theta = 45°$?
   (a) $\sin^2\theta + \cos^2\theta = 1$
   (b) $\tan^2\theta = \frac{1}{\cos^2\theta}$
   (c) $\tan^2\theta = \frac{1}{\sin^2\theta}$
   (d) $\cos^2\theta = 0$

9. Distance formula is a direct application of:
   (a) Area formula
   (b) Slope theorem
   (c) Pythagoras theorem
   (d) Projection rule

10. Which of the following is not a fundamental identity?
   (a) $\sin^2\theta + \cos^2\theta = 1$
   (b) $1 + \tan^2\theta = \sec^2\theta$
   (c) $\cot^2\theta + 1 = \csc^2\theta$
   (d) $\tan\ \theta + \cot\ \theta = 1$

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**Deductions from Fundamental Law of Trigonometry**

1. If $\sin^2\theta + \cos^2\theta = 1$, then $\sin^2\theta =$?
   (a) $1 - \cos^2\theta$
   (b) $\cos^2\theta$
   (c) $\tan^2\theta$
   (d) $\sec^2\theta - 1$

2. If $\sec^2\theta = 1 + \tan^2\theta$, what is $\tan^2\theta$?
   (a) $\sec^2\theta$
   (b) $\sec^2\theta - 1$
   (c) $\csc^2\theta - 1$
   (d) $1 + \sec^2\theta$

3. If $\csc\ \theta = 2$, what is $\sin\theta$?
   (a) $\frac{1}{2}$
   (b) 2
   (c) $\frac{1}{3}$
   (d) $\frac{3}{2}$

4. Identify obtained by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$:
   (a) $1 + \cot^2\theta = \csc^2\theta$
   (b) $\sec^2\theta = 1 + \tan^2\theta$
   (c) $\cos^2\theta = \sin^2\theta$
   (d) $\tan^2\theta = \cot^2\theta$

5. Which of the following identities is false?
   (a) $\sin^2\theta + \cos^2\theta = 1$
   (b) $\sec^2\theta - \tan^2\theta = 1$
   (c) $\cot^2\theta + 1 = \csc^2\theta$
   (d) $\tan^2\theta = 1 + \sec^2\theta$

6. Which of the following does not follow from the fundamental identity?
   (a) $\tan^2\theta + 1 = \sec^2\theta$
   (b) $\cot^2\theta + 1 = \csc^2\theta$
   (c) $\cos^2\theta = 1 - \sin^2\theta$
   (d) $\tan^2\theta = \csc^2\theta$

**Trigonometric Ratios of Allied Angles**

1. $\sin(180° - \theta) = ?$

   (a) $\sin\theta$
   (b) $-\sin\ \theta$
   (c) $\cos\theta$
   (d) $-\cos\ \theta$

2. $\cos(180° + \theta) = ?$

   (a) $\cos\theta$
   (b) $-\cos\theta$
   (c) $-\sin\theta$
   (d) $\sin\theta$

3. $\tan(270° + \theta) = ?$

(a) $-\cot\ \theta$
(b) $\cot\theta$
(c) $-\tan\ \theta$
(d) $\tan\theta$

4. $\cos\,(90^\circ - \theta) = ?$

(a) $\sin\theta$
(b) $\cos\theta$
(c) $-\sin\ \theta$
(d) $-\cos\ \theta$

5. $\tan\,(90^\circ + \theta) = ?$

(a) $\cot\theta$
(b) $-\cot\ \theta$
(c) $\tan\theta$
(d) $-\tan\ \theta$

6. Which of the following is true?
(a) $\sin\,(90^\circ + \theta) = \cos\ \theta$
(b) $\sin^2\theta = \cos^2\theta$
(c) $\cos^2\theta = \sin^2\theta$
(d) $\tan^2\theta = \cot^2\theta$

7. Which identity defines allied angles?
(a) $\sin\,(\pi - \theta) = \sin\ \theta$
(b) $\tan\,(\pi/2 + \theta) = -\cot\ \theta$
(c) Both A and B
(d) None

8. $\cos\,(270^\circ - \theta) = ?$

(a) $\sin\theta$
(b) $-\sin\ \theta$
(c) $-\cos\ \theta$
(d) $\cos\theta$

**9. Which identity is used to express $\cos^2\theta$ in terms of $\cos 2\theta$?**

(a) $\cos^2\theta = \frac{1+\cos\ 2\theta}{2}$
(b) $\cos^2\theta = \cos 2\theta$
(c) $\cos^2\theta = 1 - \sin 2\theta$
(d) None

**10. The expression $\cos^2\theta - \sin^2\theta$ equals:**

(a) $\cos^2\theta$
(b) $\cos 2\theta$
(c) $\sin\theta$
(d) $\tan\theta$

**11. In sinA sinB =?**

(a) $\frac{1}{2}[\cos(A-B) - \cos(A+B)]$
(b) $\frac{1}{2}[\cos(A+B) - \cos(A-B)]$
(c) $\sin(A+B)$
(d) $\cos(A-B)$

**12. cosA cosB =?**

(a) $\frac{1}{2}[\cos(A-B) + \cos(A+B)]$
(b) $\sin A \sin B$
(c) $\cos(A+B)$
(d) $\cos(A-B)$

**13. sinA cosB =?**

(a) $\frac{1}{2}[\sin(A+B) + \sin(A-B)]$
(b) $\cos(A+B)$
(c) $\sin(A+B)$
(d) $\cos(A-B)$

**14. Which identity helps in modulation and demodulation in signal processing?**

(a) Product-to-sum
(b) Half-angle
(c) Pythagorean
(d) Triple angle

**16. In Engineering double angle identities help simplify?**

(a) Complex integrals
(b) Oscillating systems
(c) Random signals
(d) De-vatives

**17. If $\sin^2\theta + \cos^2\theta = 1$, then this is used to:**

(a) Check vector normalization
(b) Find derivatives
(c) Simplify logarithms
(d) Eliminate constants

**18. Half-angle identities are useful in:**

(a) Calculating angles in triangles
(b) Solving differential equations
(c) Approximating trigonometric functions

(d) Both a & c

**19. The expression $\cos^2\theta - \sin^2\theta$ equals:**

(a) $\cos^2\theta$
(b) $\cos 2\theta$
(c) $\sin\theta$
(d) $\tan\theta$

**20.**Which is the result of cosxcosy using product-to-sum identity?

(a) $\frac{1}{2}[\cos(x+y) + \cos(x-y)]$
(b) $\cos\ x + \cos\ y$
(c) $\cos(x+y)$
(d) $\sin\ x \sin\ y$

**21. Product-to-sum formulas simplify:**

(a) Long expressions
(b) Integration problems
(c) Trigonometric products
(d) All √

**22. The sum-to-product identity is the reverse of:**

(a) Product-to-sum
(b) Addition formula
(c) Difference formula
(d) Half-angle formula

**23. sinA cosB is expressed as:**

(a) $\frac{1}{2}[\sin(A+B) + \sin(A-B)]$
(b) $\frac{1}{2}[\sin\ A + \cos\ B]$
(c) $\cos(A+B)$
(d) $\tan\ A \tan\ B$

**24. Which of the following does not follow the product-to-sum identity?**

(a) cosA cosB
(b) cosA sinB
(c) sinA sinB
(d) $\sin^2 A$

**25. What identity helps evaluate cos75°?**

(a) Double angle
(b) Product-to-sum
(c) Sum formula
(d) Half angle

**26. Triple angle identity helps in simplifying:**

(a) Polynomial expressions
(b) Power-reduction forms
(c) Non-linear waveforms
(d) Square roots

**27. Engineers use trigonometric identities to:**

(a) Solve signal phase equations
(b) Paint walls
(c) Estimate weights
(d) Count data

**28. Identity tan2θ =** $\frac{2\tan\theta}{1-\tan^2\theta}$ is applied in:

(a) Electrical circuit phasors
(b) Profit estimation
(c) Both a & b
(d) None of these